

17th Progress Report

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Date:	25/06/2024
Subject:	RL for Flying-V
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Outcome of the previous meeting

- Gathered and reviewed preliminary results, noted how agent actions are highly oscillatory across all algorithms.
- Discussed possibility of running algorithms on a system with rate saturated elevator. I decided will implement this saturation and rerun the algorithms on all the same flight cases but with rate-saturated elevator.

Description of the results since last meeting

- **Implemented rate saturation on elevators**, meaning now the elevator has both deflection and deflection rate limits. Used the model as shown in Figure 1.

Plots of before and after rate saturation are shown in Figure 2. From Figure 2 it can be seen that while action smoothness increases during the flight manoeuvre (from 60 s to 90 s), introducing rate saturation pushed more agents towards oscillatory policies during warmup. Interestingly, these agents with oscillatory policy from warmup seem to be fixed after the introduction of fault, most likely because changing the environment dynamics forced the agents to update, which happened to have led them to discover smoother policies.

- **Calculated the Smoothness metric.**

The smoothness of actions is also quantified for each run using the Sm formula from [1], obtained by multiplying the PSD of an agent's action during the flight manoeuvre with the corresponding frequencies. Struggled a little bit with this since there was a difficult-to-detect 'bug', where I did not properly scale the PSD obtained from FFT, meaning my smoothness values were off, and smoother actions did not have a lower Sm than rough actions.

Distribution of Sm of the IDHP(λ) algorithm before and after rate saturation are plotted in Figure 3, where it can be seen that introducing rate saturation improves the smoothness of control action, note that lower Sm means smoother time trace.

- **Performed manual hyper-parameter tuning after introducing this new rate-saturated PH-LAB**, where I change the hyper-parameters for one algorithm at a time, trying to optimize their learning performance. Arrived at the best (so far found) set of hyper-parameters for each algorithm, and gathered their metrics. As a reminder, the designed flight cases involves controlling the aircraft to follow a reference pitch signal, as shown in Figure 5, 3 faults are considered: damaged elevator, shifted CG, and damaged elevator with restricted deflection limits.

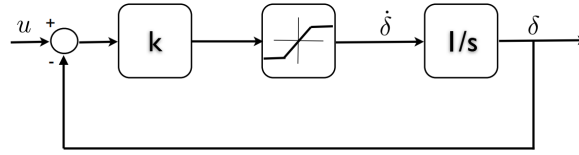


Figure 1: Actuator model with rate saturation [2], for deflection *angle* saturation I put a saturator on δ after the feedback path, just like how the CitAST model is.

The RSE during manoeuvre (from 60 s to 90 s) of all algorithms can be seen across the four flight cases in Figure 6. Currently am in the process of performing statistical testing to see if difference in performance between baseline IDHP and augmented IDHP is statistically significant, as well as their ordering. But using my eye-norm, it seems for most cases and algorithms, the augmented algorithms have lower tracking error during flight.

- **Started writing scientific article.**

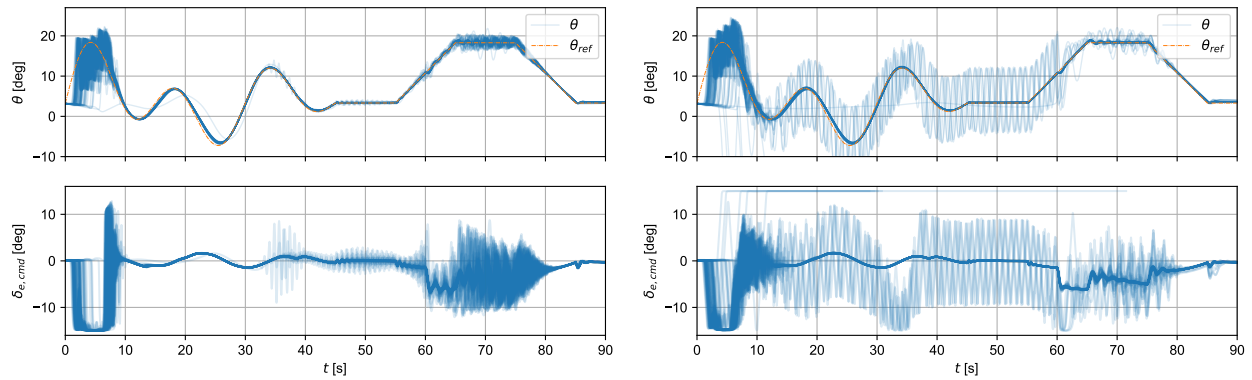
Found the AIAA paper template from the C&S bright space page.

Title i have come up with: **A Multi-step and Eligibility Trace Approach to Incremental Dual Heuristic Programming for Flight Control**

For the title + abstract, see Figure 7

Points for Discussion

- About journal, I would like to submit to a journal and would like to talk about it.
 - Sub point, is it possible to submit a paper to a conference and then to a journal?
- What should I be thinking about when the thesis rubric says scientific significance. Alternatively, what is scientific significance.
- Normalized errors, how does one do normalization and what is the advantage to using them versus un-normalized errors.



(a) Without rate saturation.

(b) With rate saturation

Figure 2: Monte Carlo runs of IDHP(λ) with and without rate saturation.

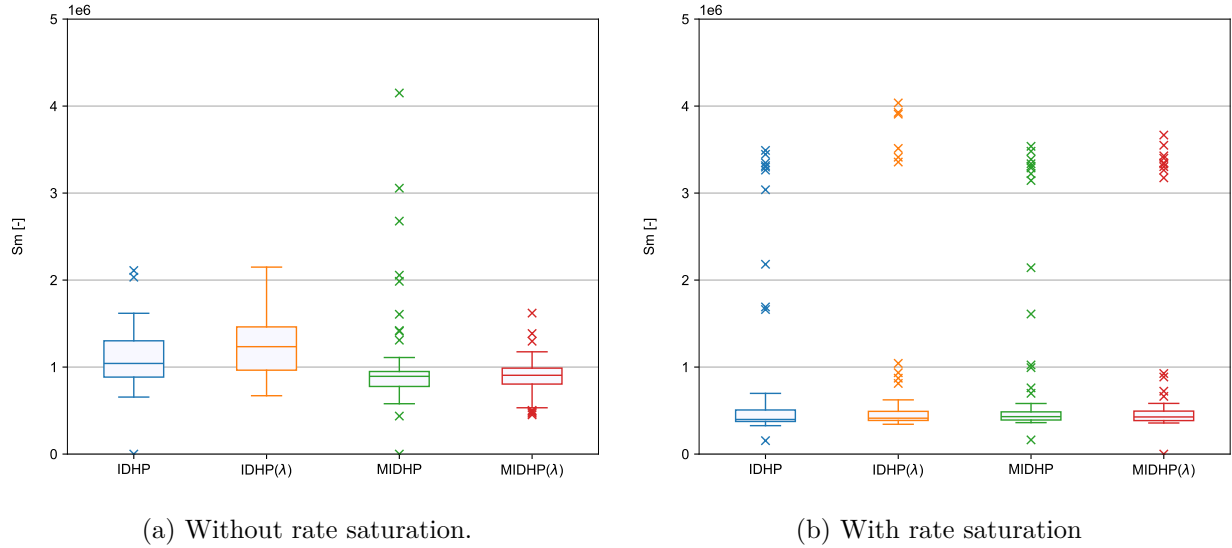


Figure 3: Smoothness values for the elevator action from the IDHP(λ) Monte Carlo results with and without rate saturation.

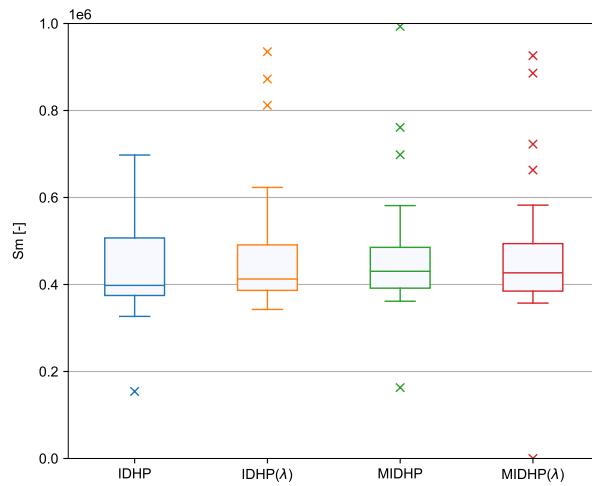


Figure 4: Smoothness value distribution from Figure 3b.

Plans for next 2 weeks

1. Continue refining plots (open for suggestions).
2. Continue writing paper, finish up introduction. Then work on the sections of the paper in the order they appear. After intro i will work on Background, then onto Methodlogy, then Results and Discussion, then Conclusion.

References

- [1] Siddharth Mysore, Bassel Mabsout, Renato Mancuso, and Kate Saenko. “Regularizing action policies for smooth control with reinforcement learning”. In: *2021 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE. 2021, pp. 1810–1816.
- [2] Yildiray Yildiz and Ilya Kolmanovsky. “Stability properties and cross-coupling performance of the control allocation scheme CAPIO”. In: *Journal of Guidance, Control, and Dynamics* 34.4 (2011), pp. 1190–1196.

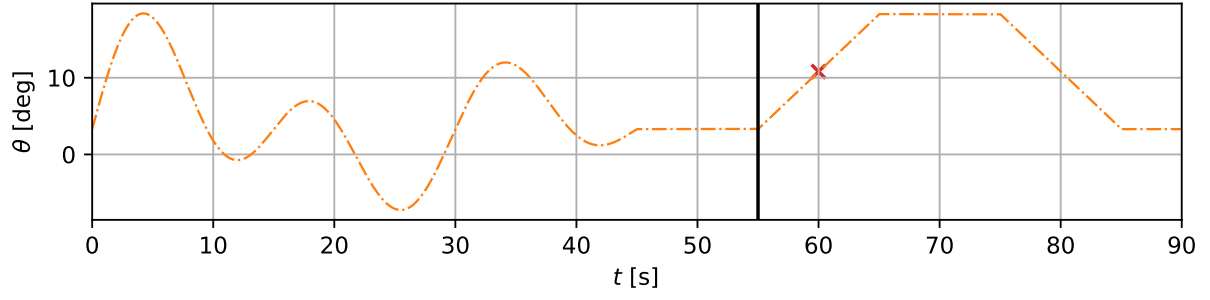
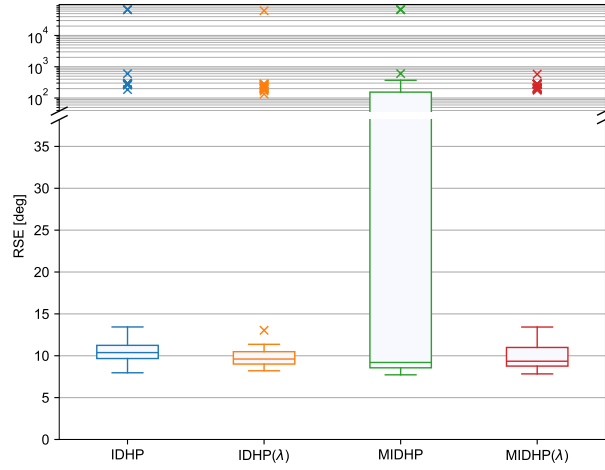
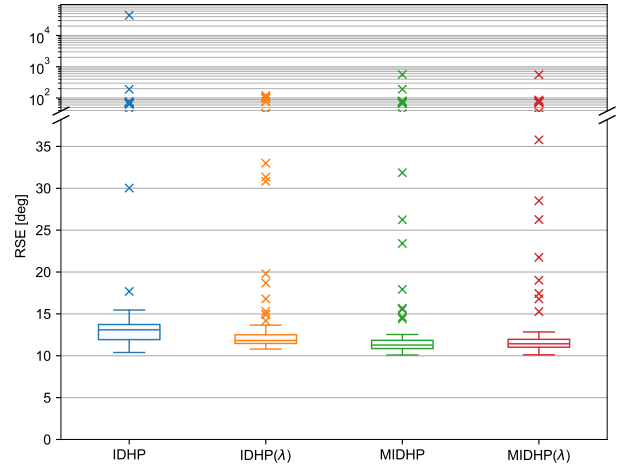


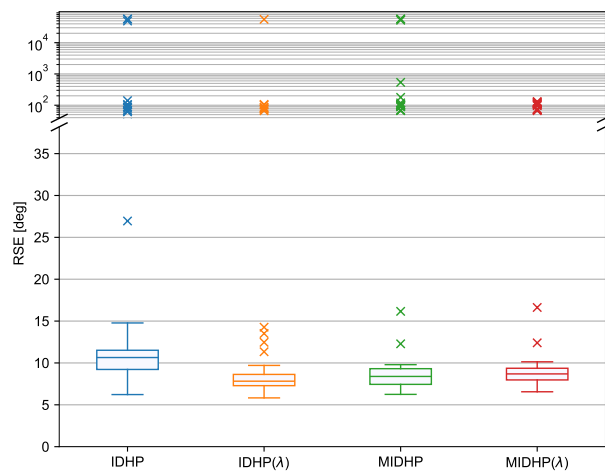
Figure 5: Reference pitch signal, the period before the black line will be referred to as the warmup phase, and afterwards is the inflight phase, the red cross at 60 s marks where faults will be triggered.



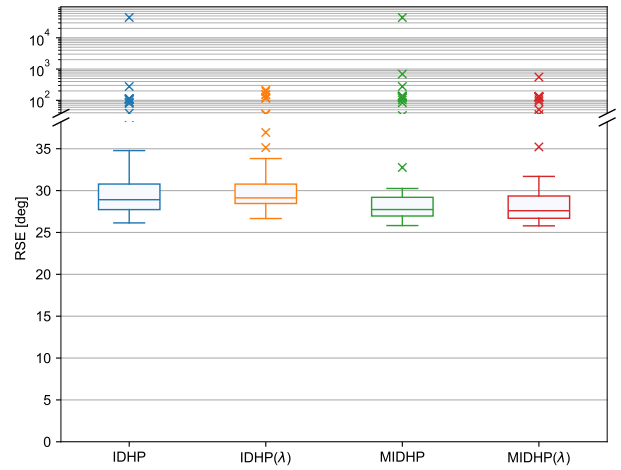
(a) No fault.



(b) Damaged elevator.



(c) Shifted CG.



(d) Damaged elevator and restricted elevator deflection limits.

Figure 6: Boxplots showing RSE in-flight (from 60 s to 90 s) of all four algorithms on each of the four flight cases.

A Multi-step and Eligibility Trace Approach to Incremental Dual Heuristic Programming for Flight Control

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Incremental Dual Heuristic Programming (IDHP) is a successor to the Dual Heuristic Programming (DHP) algorithm that uses an incremental system model, this algorithm showed promising flight control performance and tolerance of faults in simulation experiments. This paper studies the potential for extending IDHP through augmenting the computation of agent updates and returns, more specifically, by using eligibility trace updates and multi-step temporal difference error. This results in the IDHP(λ) and MIDHP(λ) algorithms, which are compared against IDHP in several simulated flight control scenarios with faults introduced mid-flight. The results demonstrate that the proposed algorithms have improved flight control performance and fault tolerance in terms of tracking errors when controlling a nominal aircraft and an aircraft with faults introduced.

Nomenclature

(Nomenclature entries should have the units identified)

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Figure 7: Draft paper front page.